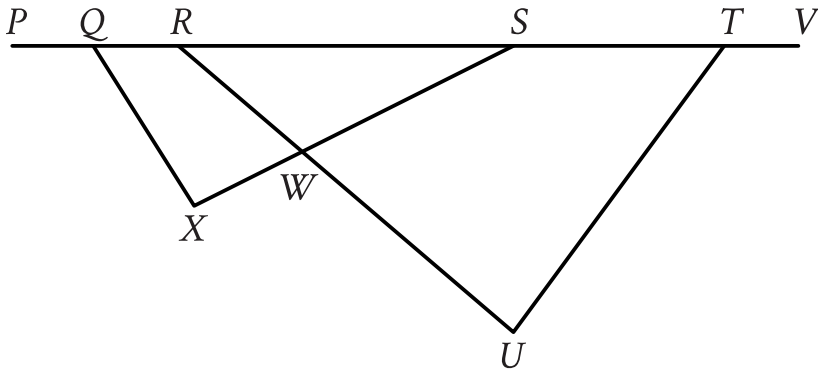
## Question ID: e10d8313

**Question**



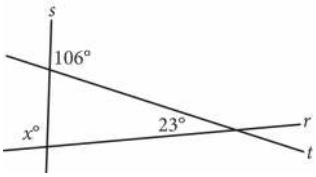

Note: Figure not drawn to scale.

In the figure shown, points $Q$, $R$, $S$, and $T$ lie on line segment $PV$, and line segment $RU$ intersects line segment $SX$ at point $W$. The measure of $\angle SQX$ is $48°$, the measure of $\angle SXQ$ is $86°$, the measure of $\angle SWU$ is $85°$, and the measure of $\angle VTU$ is $162°$. What is the measure, in degrees, of $\angle TUR$?

# Question ID: f88f27e5

**Question**

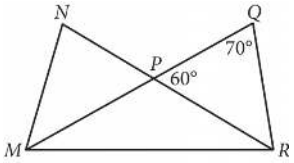
Intersecting lines r, s, and t are shown below.




What is the value of x ?

# Question ID: 947a3cde

**Question**




In the figure above, $\overline{MQ}$ and $\overline{NR}$ intersect at point P, $NP = QP$, and $MP = PR$. What is the measure, in degrees, of $\angle QMR$ ? (Disregard the degree symbol when gridding your answer.)

# Question ID: f7dbde16

**Question**

In triangles $LMN$ and $RST$, angles $L$ and $R$ each have measure $60°$, $LN = 10$, and $RT = 30$. Which additional piece of information is sufficient to prove that triangle $LMN$ is similar to triangle $RST$?

**Answer**

A. $MN = 7$ and $ST = 7$

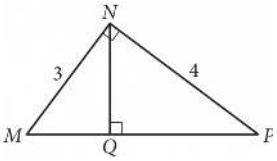
B. $MN = 7$ and $ST = 21$

C. The measures of angles $M$ and $S$ are $70°$ and $60°$, respectively.

D. The measures of angles $M$ and $T$ are $70°$ and $50°$, respectively.

# Question ID: 740bf79f

**Question**




In the figure above, what is the length of $\overline{NQ}$ ?

**Answer**

A. 2.2

B. 2.3

C. 2.4

D. 2.5

# Question ID: fecc446d

**Question**

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)°$. The sum of the measures of four of the eight angles is $k°$. Which of the following could NOT be equivalent to $k$, for all values of $x$?

**Answer**

A. $-14x + 1{,}540$

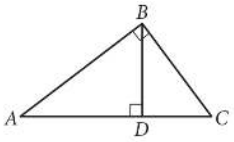
B. $14x - 320$

C. $-28x + 1{,}720$

D. $360$

# Question ID: 6a3fbec3

**Question**

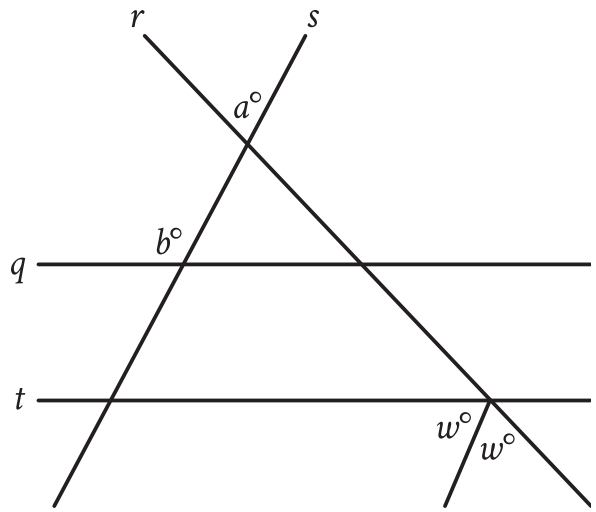



Note: Figure not drawn to scale.

In the figure above, $BD = 6$ and $AD = 8$. What is the length of $\overline{DC}$ ?

## Question ID: 17912810

**Question**




Note: Figure not drawn to scale.

In the figure, parallel lines $q$ and $t$ are intersected by lines $r$ and $s$. If $a = 43$ and $b = 122$, what is the value of $w$?

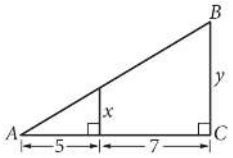
# Question ID: 5b4757df

**Question**

In triangle $RST$, angle $T$ is a right angle, point $L$ lies on $\overline{RS}$, point $K$ lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is $72$ units, the length of $\overline{LK}$ is $24$ units, and the area of triangle $RST$ is $792$ square units, what is the length of $\overline{KT}$, in units?

# Question ID: eeb4143c

**Question**




Note: Figure not drawn to scale.

The area of triangle ABC above is at least 48 but no more than 60. If y is an integer, what is one possible value of x ?

# Question ID: 322a6dfe

**Question**

Quadrilaterals $PQRS$ and $WXYZ$ are similar, where $P$, $Q$, and $R$ correspond to $W$, $X$, and $Y$, respectively. The measure of $\angle S$ is $135°$, $PS = 45$, and $WZ = 9$. What is the measure of $\angle Z$?

**Answer**

A. $5°$

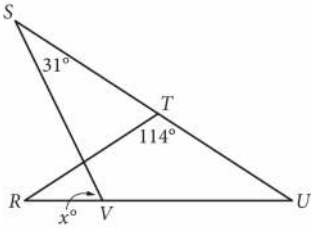
B. $27°$

C. $45°$

D. $135°$

# Question ID: bd7f6e30

**Question**




In the figure above, $RT = TU$. What is the value of x ?

**Answer**

A. 72

B. 66

C. 64

D. 58